A Smooth Non-Euclidean Counterexample to the Strong Birkhoff–James Basis Conjecture

A full counterexample to Conjecture 2.11 of arXiv:1407.5016

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Abstract

We disprove the conjecture that a finite-dimensional real smooth normed space of dimension at least three is Euclidean whenever every unit vector belongs to a strongly orthonormal Hamel basis in the sense of Birkhoff–James. An explicit smooth, strictly convex, non-Euclidean example exists in every such dimension. It is the Euclidean direct sum of a non-Euclidean Radon plane and a Euclidean space. The required basis through each unit vector is given by a closed formula.

1 Source conjecture and answer

Sain–Paul–Debnath ask in Conjecture 2.11 (page 6) whether the following property characterizes inner-product spaces among finite-dimensional real smooth normed spaces of dimension at least three: every point of the unit sphere is contained in a strongly orthonormal Hamel basis in the sense of Birkhoff–James.

> We conclude with the following conjecture:
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> **Conjecture 2.11.** *A finite dimensional(≥ 3) real smooth normed linear space is an inner product space iff given any element on the unit sphere of the space there exists a strongly orthonormal Hamel basis in the sense of Birkhoff-James containing that element.*

The answer is no, even if strict convexity is added. We first construct an explicit non-Euclidean Radon plane and then give counterexamples in all dimensions $n \geq 3$.

2 Idea of Proof

In a smooth space, $x \perp_B y$ exactly when the unique norming functional at x annihilates y. In a strictly convex space this already implies the strict inequality defining $x \perp_{SB} y$. A Radon plane is a normed plane in which Birkhoff–James orthogonality is symmetric. Thus every unit vector a in such a plane has a unit partner b with mutual orthogonality.

Now take the Euclidean direct sum of that plane and $\mathbb{R}^{n-2}$. An arbitrary unit vector has the form (ra, sc) with $r^2 + s^2 = 1$. The vectors (ra, sc), $(-sa, rc)$, and $(b, 0)$, followed by an orthonormal completion of c in the Euclidean summand, form the desired basis. Their norming functionals have the same two-dimensional rotation pattern, so every off-diagonal pairing vanishes.

3 An explicit smooth non-Euclidean Radon plane

Fix $1 < p < \infty$, $p \neq 2$, and let $q = p/(p-1)$. Define N_p on $\mathbb{R}^2$ by

$$N_p(x, y) = \begin{cases} (|x|^p + |y|^p)^{1/p}, & xy \geq 0, \\ (|x|^q + |y|^q)^{1/q}, & xy \leq 0. \end{cases} \tag{1}$$

The definitions agree on the coordinate axes. This mixed-exponent Radon construction is standard and is also recorded in the source TeX accompanying the authors' related paper [2]; the Euclidean-sum lifting below is the step that defeats the higher-dimensional conjecture. We include all details needed here.

Lemma 1. *The function N_p is a smooth, strictly convex norm. It is non-Euclidean, and Birkhoff–James orthogonality in $E_p = (\mathbb{R}^2, N_p)$ is symmetric.*

Proof. The upper boundary of the proposed unit ball is the graph

$$f(x) = \begin{cases} (1 - |x|^q)^{1/q}, & -1 \leq x \leq 0, \\ (1 - x^p)^{1/p}, & 0 \leq x \leq 1. \end{cases}$$

Each piece is strictly concave, and their one-sided derivatives at zero are both zero. Hence f is strictly concave. The lower boundary is $-f(-x)$, so the enclosed centrally symmetric set is convex and strictly convex. Formula (1) is its Minkowski functional and therefore a norm. The two arcs have matching horizontal tangents at $(0, \pm 1)$ and matching vertical tangents at $(\pm 1, 0)$; elsewhere they are smooth. Thus every nonzero point has a unique norming functional.

Let $a = (a_1, a_2)$ be a unit vector off the axes and write j_a for its unique norming functional, identified with its coefficient vector. If $a_1 a_2 > 0$, then

$$j_a = (\operatorname{sgn}(a_1)|a_1|^{p-1}, \operatorname{sgn}(a_2)|a_2|^{p-1}).$$

Put $b = (-(j_a)_2, (j_a)_1)$. Its coordinates have opposite signs and, using $(p-1)q = p$,

$$N_p(b)^q = |(j_a)_2|^q + |(j_a)_1|^q = |a_2|^p + |a_1|^p = 1.$$

Moreover its norming functional is $j_b = (-a_2, a_1)$, whence $j_a(b) = j_b(a) = 0$. If $a_1 a_2 < 0$, the same calculation interchanges p and q and uses $(q-1)p = q$. The axis cases are immediate. Since in a smooth normed space $u \perp_B v$ iff $j_u(v) = 0$, orthogonality is symmetric.

Finally, $N_p(1, 1) = 2^{1/p}$ whereas $N_p(1, -1) = 2^{1/q}$. The parallelogram identity would give $2^{2/p} + 2^{2/q} = 4$, which, by strict convexity of $t \mapsto 2^t$ and $2/p + 2/q = 2$, holds only for $p = q = 2$. Thus E_p is not Euclidean. □

4 The counterexample in every dimension

We use the following elementary smoothness observation.

Lemma 2. *Let X be smooth and strictly convex, and let j_x denote the unique norming functional of $x \in S_X$. Unit vectors $e_1, \ldots, e_n$ form a strongly orthonormal Hamel basis in the sense of Birkhoff–James if*

$$j_{e_i}(e_k) = \delta_{ik} \qquad (1 \leq i, k \leq n).$$

Proof. For w in the span of all e_k with $k \neq i$, the displayed identities give $j_{e_i}(w) = 0$, hence $e_i \perp_B w$. Therefore $\|e_i + w\| \geq 1$. Equality for nonzero w would put two distinct points e_i and $e_i + w$ on the unit sphere in the supporting hyperplane $j_{e_i} = 1$, contradicting strict convexity. Hence $\|e_i + w\| > 1$ whenever $w \neq 0$. □

Theorem 3. *For every integer $n \geq 3$ there is a smooth, strictly convex, non-inner-product normed space X_n in which every unit vector belongs to a strongly orthonormal Hamel basis in the sense of Birkhoff–James. Consequently Conjecture 2.11 of Sain–Paul–Debnath is false.*

Proof. Let $H = \ell_2^{n-2}$ and equip

$$X_n = E_p \oplus_2 H, \qquad \|(u,h)\|_X = (N_p(u)^2 + \|h\|_2^2)^{1/2}.$$

This norm is smooth and strictly convex because both summands are. It is not an inner-product norm, since its restriction to $E_p \oplus \{0\}$ is N_p.

Take $x = (u,h) \in S_{X_n}$ and put $r = N_p(u)$ and $s = \|h\|_2$, so $r^2 + s^2 = 1$. If $r > 0$, write $u = ra$ with $a \in S_{E_p}$; if $r = 0$, choose any $a \in S_{E_p}$. By Lemma 1, choose $b \in S_{E_p}$ with $j_a(b) = j_b(a) = 0$. If $s > 0$, write $h = sc_1$ with $c_1 \in S_H$; if $s = 0$, choose any $c_1 \in S_H$. Complete c_1 to a Euclidean orthonormal basis $c_1, \ldots, c_{n-2}$ of H.

Define

$$\begin{aligned} e_1 &= (ra, sc_1) = x, \qquad & e_2 &= (-sa, rc_1), \qquad & e_3 &= (b, 0), \\ e_{k+2} &= (0, c_k) & & (2 \leq k \leq n-2). \end{aligned}$$

The last line is absent for $n = 3$. These are unit vectors and form a basis: e_1, e_2 are an orthogonal two-by-two change of coordinates in $\operatorname{span}\{(a,0),(0,c_1)\}$, while the remaining displayed vectors complete the two summands.

For a unit vector (v,z) in the Euclidean sum, its norming functional is

$$j_{(v,z)} = \big(N_p(v) j_{v/N_p(v)},\ z\big),$$

with first component zero when $v = 0$. Consequently

$$j_{e_1} = (rj_a, sc_1), \quad j_{e_2} = (-sj_a, rc_1), \quad j_{e_3} = (j_b, 0), \quad j_{e_{k+2}} = (0, c_k),$$

where vectors in H are identified with their Euclidean functionals. The relations $j_a(b) = j_b(a) = 0$, $r^2 + s^2 = 1$, and $\langle c_i, c_k \rangle = \delta_{ik}$ now give

$$j_{e_i}(e_k) = \delta_{ik}$$

for every pair of basis indices. Lemma 2 proves that this is a strongly orthonormal Hamel basis containing the prescribed x. □

5 Verification and novelty scope

The accompanying deterministic script checks the mixed norm, its norming functionals, the mutual Radon identities, and the full biorthogonality matrix for many nondegenerate and edge-case vectors in dimensions three through seven. It also records the explicit failure of the parallelogram law.

A bounded search of the run indexes and exact terminology found the source paper and subsequent general work on strongly orthonormal/Auerbach bases, but no paper resolving Conjecture 2.11 by the Radon-plane Euclidean-sum construction. The novelty claim is limited to this explicit counterexample and proof.

References

[1] D. Sain, K. Paul, and L. Debnath, *Characterization of inner product spaces*, arXiv:1407.5016 (2014), Conjecture 2.11.

[2] D. Sain, K. Paul, and K. Jha, *Strictly convex space: Strong orthogonality and conjugate diameters*, arXiv:1411.1464 (2014).